

(a) First we compute C . We work in spherical coordinates, and since we only integrate wrt r , the volume differential dV can be expressed as $4\pi r^2 dr$. Integrating along r gives

$$Q = -e = \int_0^\infty \rho(r) dV = \int_0^\infty -C4\pi r^2 \exp\left(-\frac{2r}{a_0}\right) dr = -C4\pi \int_0^\infty r^2 \exp\left(-\frac{2r}{a_0}\right) dr.$$

We compute the integral. Using integration by parts, we see

$$\begin{aligned} \int r^2 \exp\left(-\frac{2r}{a_0}\right) dr &= r^2 \left(-\frac{a_0}{2}\right) \exp\left(-\frac{2r}{a_0}\right) - \int 2r \left(-\frac{a_0}{2}\right) \exp\left(-\frac{2r}{a_0}\right) dr \\ &= -\frac{r^2 a_0}{2} \exp\left(-\frac{2r}{a_0}\right) + a_0 \int r \exp\left(-\frac{2r}{a_0}\right) dr. \end{aligned}$$

We use integration by parts on the new integral:

$$\begin{aligned} \int r \exp\left(-\frac{2r}{a_0}\right) dr &= r \left(-\frac{a_0}{2}\right) \exp\left(-\frac{2r}{a_0}\right) - \int -\frac{a_0}{2} \exp\left(-\frac{2r}{a_0}\right) dr \\ &= -\frac{ra_0}{2} \exp\left(-\frac{2r}{a_0}\right) + \frac{a_0}{2} \int \exp\left(-\frac{2r}{a_0}\right) dr = -\frac{ra_0}{2} \exp\left(-\frac{2r}{a_0}\right) + \frac{a_0}{2} \left(-\frac{a_0}{2}\right) \exp\left(-\frac{2r}{a_0}\right) \\ &= \left(-\frac{ra_0}{2} - \frac{a_0^2}{4}\right) \exp\left(-\frac{2r}{a_0}\right). \end{aligned}$$

We plug this back into the original integral

$$\begin{aligned} \int r^2 \exp\left(-\frac{2r}{a_0}\right) dr &= -\frac{r^2 a_0}{2} \exp\left(-\frac{2r}{a_0}\right) + a_0 \left(-\frac{ra_0}{2} - \frac{a_0^2}{4}\right) \exp\left(-\frac{2r}{a_0}\right) \\ &= -\left(\frac{r^2 a_0}{2} + \frac{ra_0^2}{2} + \frac{a_0^3}{4}\right) \exp\left(-\frac{2r}{a_0}\right). \end{aligned}$$

We evaluate r from 0 to ∞ :

$$-\left(\frac{r^2 a_0}{2} + \frac{ra_0^2}{2} + \frac{a_0^3}{4}\right) \exp\left(-\frac{2r}{a_0}\right) \Big|_0^\infty = \frac{a_0^3}{4},$$

since the exponential grows faster than the polynomials, meaning the limit $r \rightarrow \infty$ is 0.

Now we go back to our expression for C :

$$-e = -C4\pi \frac{a_0^3}{4} \implies C = \frac{e}{a_0^3 \pi}.$$

To now compute Q in a sphere of radius a_0 , we first compute Q_{cloud} by computing the above integral from 0 to a_0 :

$$\begin{aligned} \int_0^{a_0} r^2 \exp\left(-\frac{2r}{a_0}\right) dr &= -\left(\frac{r^2 a_0}{2} + \frac{ra_0^2}{2} + \frac{a_0^3}{4}\right) \exp\left(-\frac{2r}{a_0}\right) \Big|_0^{a_0} \\ &= -\left(\frac{a_0^3}{2} + \frac{a_0^3}{2} + \frac{a_0^3}{4}\right) \exp(-2) + \frac{a_0^3}{4} = -e^{-2} \frac{5a_0^3}{4} + \frac{a_0^3}{4} = \frac{a_0^3}{4} (1 - 5e^{-2}). \end{aligned}$$

Note that Q_{cloud} is not equal to this integral, but rather

$$Q_{\text{cloud}} = \int_0^{a_0} \rho(r) \, dV = -C4\pi \frac{a_0^3}{4} (1 - 5e^{-2}) = -e (1 - 5e^{-2}).$$

Overall Q requires us to add the proton's contribution, giving

$$Q = e - e (1 - 5e^{-2}) = e - e + 5e^{-2}e = 5e^{-2}e,$$

where we note that Euler's number and the electron charge annoyingly have the same symbol. Taking $e = 1.602 \cdot 10^{-19}$ Coulombs, this evaluates to $Q \approx 1.08404 \cdot 10^{-19}$ Coulombs.

(b) We construct a sphere of radius a_0 as our Gaussian surface and apply Gauss' law

$$\int \mathbf{E} \cdot d\mathbf{A} = \frac{Q}{\epsilon_0}.$$

Of course by the spherical symmetry of $\rho(r)$ and the field, the integral on the left is simply E times the surface area

$$\int \mathbf{E} \cdot d\mathbf{A} = E(4\pi a_0^2) = \frac{Q}{\epsilon_0}.$$

Plugging Q into this gives

$$E4\pi a_0^2 = \frac{5e}{e^2\epsilon_0} \implies E = \frac{5e}{4\pi a_0^2 e^2 \epsilon_0}.$$

Using $a_0 \approx 0.53 \cdot 10^{-10}$ and $\epsilon_0 \approx 8.85418782 \cdot 10^{-12}$ gives that numerically, we have

$$E \approx (1.08404 \cdot 10^{-19}) \frac{1}{4\pi(0.53 \cdot 10^{-10})^2 (8.85418782 \cdot 10^{-12})} \approx 3.46845 \cdot 10^{11} \text{ N/C}.$$